

PSCAD V5.0.2 Three-Tier Controller Implementation Guide

Project: PCS3440 + DELTERR Mi2 BESS Microgrid Controller

Platform: PSCAD/EMTDC v5.0.2

System: 2 × 5 MW / 10 MWh BESS (10 MW / 20 MWh aggregate)

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Table of Contents

1. [Deliverables Overview](#)
 2. [File Structure](#)
 3. [Quick-Start Integration Guide](#)
 4. [Unit Controller \(UC\) Implementation](#)
 5. [Microgrid Controller \(MGC\) Implementation](#)
 6. [EMS/SCADA Implementation](#)
 7. [System Integration](#)
 8. [Signal Routing & Data Labels](#)
 9. [Multi-Rate Configuration](#)
 10. [Test Procedures](#)
 11. [Troubleshooting](#)
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1. Deliverables Overview

1.1 Generated Files

File	Description	Usage
<code>BESS_3Tier_Controller.pscx</code>	Integrated project with all three controller tiers, component instances, data label routing, and monitoring outputs	Primary project file — open in PSCAD V5.0.2
<code>Subsystems/UC_Controller.pscx</code>	Unit Controller standalone module	Import as library component; or use as reference
<code>Subsystems/MGC_Controller.pscx</code>	Microgrid Controller standalone module	Import as library component; or use as reference
<code>Subsystems/EMS_SCADA.pscx</code>	EMS/SCADA Dispatch standalone module	Import as library component; or use as reference
<code>Resources/uc_controller.f</code>	UC Fortran subroutine (ALG-01 to ALG-05)	Drop into UC page as Fortran Component
<code>Resources/mgc_controller.f</code>	MGC Fortran subroutine (ALG-06 to ALG-10)	Drop into MGC page as Fortran Component
<code>Resources/ems_scada.f</code>	EMS Fortran subroutine (ALG-11 to ALG-14)	Drop into EMS page as Fortran Component
<code>Resources/dispatch_schedule.txt</code>	24-hour dispatch schedule (TOU + peak-shaving)	Referenced by EMS File Read component
<code>generate_pscx.py</code>	Python generator script	Re-generate or modify PSCX files

1.2 Implementation Approaches

The implementation provides **two parallel approaches** — use whichever fits your workflow:

Approach	Components	Pros	Cons
A: Graphical blocks	PSCX files with comparators, integrators, selectors, logic gates, flip-flops	Visual, debuggable in PSCAD GUI, no compilation needed	More wiring work, harder to modify algorithms
B: Fortran sub-routines	.f files compiled as FileCmp components	Compact, easy to modify logic, better for complex state machines	Requires Fortran compilation, harder to debug visually

Recommended workflow: Use graphical blocks (Approach A) first to validate the concept, then migrate to Fortran components (Approach B) for production.

2. File Structure

```

PSCAD_Controller_Implementation/
├── BESS_3Tier_Controller.pscx      ← Main integrated project
├── generate_pscx.py                ← Generator script
├── Subsystems/
│   ├── UC_Controller.pscx         ← Unit Controller module
│   ├── MGC_Controller.pscx       ← Microgrid Controller module
│   └── EMS_SCADA.pscx            ← EMS/SCADA module
├── Resources/
│   ├── uc_controller.f            ← UC Fortran code
│   ├── mgc_controller.f          ← MGC Fortran code
│   ├── ems_scada.f               ← EMS Fortran code
│   └── dispatch_schedule.txt     ← 24-hour TOU schedule

```

3. Quick-Start Integration Guide

Step-by-Step: Adding Controllers to Your Existing PCS_Unit_1 Project

Step 1: Backup

Copy PCS_Unit_1.pscx → PCS_Unit_1_backup.pscx

Step 2: Import Controller Definitions

1. Open PCS_Unit_1.pscx in PSCAD V5.0.2
2. **Right-click** in the Definitions panel → **Import Definition**
3. Navigate to Subsystems/UC_Controller.pscx and import UC_Controller
4. Repeat for MGC_Controller and EMS_SCADA

Step 3: Create New Page Modules

For each controller, create a new Page Module in the project:

Unit Controller 1 (UC_1):

1. Right-click the project tree → **Add Page Module**
2. Name: UC_1
3. Double-click the new page → open properties
4. Set **Multi-Rate Factor = 20** (execution every 1 ms with $\Delta t = 50\mu s$)
5. Drag UC_Controller component onto the page canvas
6. Connect data labels per §8

Unit Controller 2 (UC_2):

- Same as UC_1, name it UC_2, Multi-Rate Factor = 20

Microgrid Controller (MGC):

1. Add Page Module, name: MGC
2. Set **Multi-Rate Factor = 200** (execution every 10 ms)
3. Place MGC_Controller component

EMS/SCADA (EMS):

1. Add Page Module, name: EMS
2. Set **Multi-Rate Factor = 20000** (execution every 1 s)
3. Place EMS_SCADA component

Step 4: Connect Data Labels

Use the naming convention from §8 to create data labels that route signals between pages:

Source Page → Data Label (Export) → Target Page → Data Label (Import)

Example:

- On EMS page: Create output data label EMS_MGC_P_dispatch connected to EMS P_dispatch output
- On MGC page: Create input data label EMS_MGC_P_dispatch connected to MGC P_dispatch input

Step 5: Connect to PCS+BESS Black Box

On the Main page or UC pages, connect UC outputs to BESS_1_1 inputs:

UC Output	PCS Connection	Method
P_ref	PQref dimension 1	Via datamerge (already exists in Main)
Q_ref	PQref dimension 2	Via datamerge (already exists in Main)
MCB_DRV	MCB_DRV1 breaker control	Direct wire or data label
AVR_set	AVR slider	Slider control signal
FW_EN_set	FW_EN slider	Slider control signal
VRT_Enable_set	VRT_Enable slider	Slider control signal
PROT_Enable_set	PROT_Enable slider	Slider control signal

Critical: The existing `Pset` and `Qset` slider controls on the Main page must be disconnected or bypassed — the UC now generates `P_ref` and `Q_ref`.

Step 6: Extend Simulation Duration

Edit project settings:

- **Time Duration:** Change from 0.5s to **25.0s** (for full 24-hour cycle at 1s/hour compression)
- **Time Step:** Keep at 50 μ s
- **Sample Step:** Keep at 200 μ s

Step 7: Build and Run

1. **Build** the project (Ctrl+B)
2. Verify no compilation errors
3. **Run** simulation
4. Monitor outputs on the graph frames placed within each controller page

4. Unit Controller (UC) Implementation

4.1 PSCX Block Architecture

The UC is implemented as a `UserCmpDefn` named `UC_Controller` with the following internal sections:

UC_Controller Page Module (Multi-Rate Factor = 20)

Section 1: RAMP-RATE LIMITER

- └─ Input: P_cmd, Ramp_limit
- └─ Summer ($\Delta = P_{\text{cmd}} - P_{\text{ref_internal}}$)
- └─ Hard Limiter ($\pm \Delta_{\text{max}}$ based on ramp rate)
- └─ Integrator (accumulates ramped output)
- └─ Output: P_ref_ramped

Section 2: SOC PROTECTION (DERATING)

- └─ Input: SOC, P_ref_ramped
- └─ Comparators: SOC vs FLOOR(10%), LOW_WARN(15%), HIGH_WARN(90%), CEILING(95%)
- └─ Derating Gain: $(\text{SOC} - \text{FLOOR}) / (\text{LOW_WARN} - \text{FLOOR}) \times 0.2$
- └─ Multiplier: P_ramped $\times$ derate_factor
- └─ Selector: zone-based output selection
- └─ Output: P_target_clamped, CLAMP_ACTIVE

Section 3: FAULT HANDLING

- └─ Input: PCS_Fault[4], V_dc, MGC_heartbeat
- └─ OR gate: any PCS fault
- └─ Comparators: $V_{\text{dc}} > 1450\text{V}$, $V_{\text{dc}} < 1200\text{V}$
- └─ Timer + Comparator: heartbeat timeout (500ms)
- └─ SR Flip-Flops: fault latching (PCS, DC_OV, DC_UV, COMM_LOSS)
- └─ OR gate: combine all fault bits
- └─ Output: UC_Fault, HARD_FAULT

Section 4: STATE MACHINE

- └─ Input: Enable, Mode_cmd, HARD_FAULT, MCB_status, DC_OK
- └─ 7 $\times$ SR Flip-Flops (one per state)
- └─ AND gates: transition conditions
- └─ OR gates: reset conditions
- └─ Time Delay: fault clear (5s)
- └─ Output: UC_State (encoded), ST_RUNNING flag

Section 5: MODE CONFIGURATION

- └─ Input: Mode_cmd
- └─ Selector blocks (5-input): one per slider output
- └─ Output: AVR_out, FW_EN_out, VRT_EN_out, PROT_EN_out

Section 6: AVAILABLE POWER CALCULATION

- └─ Input: SOC
- └─ Comparators + gain: linear derating calculation
- └─ Selector: zone selection
- └─ Output: P_avail, P_chg_avail

Section 7: OUTPUT ASSEMBLY + MONITORING

- └─ Multiplier gates (output $\times$ RUNNING flag)
- └─ Heartbeat counter
- └─ Graph channels: P_ref, SOC, State, Faults

4.2 Port Table

Inputs (Left Side of Symbol — Blue)

Port	Name	Type	Dim	From
1	P_meas	Real	1	PCS FFT_Mod_2
2	Q_meas	Real	1	PCS FFT_Mod_2
3	SOC	Real	1	BESS model
4	V_bat	Real	1	BESS model
5	V_dc	Real	1	DC link
6	V_ac_rms	Real	3	FFT_Mod_2
7	PCS_Fault	Int	4	Ctrl_1_2
8	MCB_status	Int	1	Breaker
9	P_cmd	Real	1	MGC
10	Q_cmd	Real	1	MGC
11	Mode_cmd	Int	1	MGC
12	Enable	Int	1	MGC
13	Ramp_limit	Real	1	MGC
14	MGC_heartbeat	Int	1	MGC

Outputs (Right Side of Symbol — Blue)

Port	Name	Type	Dim	To
15	P_ref	Real	1	PCS PQref(1)
16	Q_ref	Real	1	PCS PQref(2)
17	MCB_DRV	Int	1	PCS breaker
18	AVR_set	Int	1	PCS slider
19	FW_EN_set	Int	1	PCS slider
20	VRT_Enable_set	Int	1	PCS slider
21	PROT_Enable_set	Int	1	PCS slider
22	UC_Status	Int	1	MGC
23	UC_Fault	Int	1	MGC
24	UC_Mode	Int	1	MGC
25	P_available	Real	1	MGC
26	P_charge_avail	Real	1	MGC
27	SOC_out	Real	1	MGC
28	P_meas_out	Real	1	MGC
29	UC_heartbeat	Int	1	MGC

4.3 Configurable Parameters

Double-click the UC component instance to access these parameters:

Category	Parameter	Default	Unit	Description
Ramp Rate	DEFAULT_RAMP	833.33	kW/s	Default ramp rate per unit
	MIN_RAMP	100	kW/s	Minimum allowed ramp rate
	MAX_RAMP	50000	kW/s	Maximum allowed ramp rate
SOC Protection	SOC_FLOOR	10.0	%	Hard discharge cutoff
	SOC_CEILING	95.0	%	Hard charge cutoff
	SOC_LOW_WARN	15.0	%	Discharge derating begins
	SOC_HIGH_WARN	90.0	%	Charge derating begins
	P_RATED	5000	kW	Rated power per unit
Faults	HB_TIMEOUT	0.5	s	MGC heartbeat timeout
	FAULT_CLEAR_DELAY	5.0	s	Post-fault cool-down
	VDC_OV	1450	V	DC overvoltage threshold
	VDC_UV	1200	V	DC undervoltage threshold
	VDC_NOM	1332	V	DC nominal voltage

4.4 Fortran Alternative

To use the Fortran implementation instead of graphical blocks:

1. Create a new page module UC_1 with Multi-Rate Factor = 20
2. Add a **Fortran Component** (FileCmp) pointing to Resources/uc_controller.f
3. Connect input/output data labels to the subroutine arguments
4. The subroutine signature is:

5. Microgrid Controller (MGC) Implementation

5.1 PSCX Block Architecture

MGC_Controller Page Module (Multi-Rate Factor = 200)

- Section 1: SOC-WEIGHTED POWER SHARING
 - Inputs: P_total, SOC_1, SOC_2, UC1_ok, UC2_ok
 - Summer: SOC_sum = SOC_1 + SOC_2
 - Dividers: weight_1 = SOC_1/SOC_sum, weight_2 = SOC_2/SOC_sum
 - Multipliers: P_cmd_i = P_total × weight_i
 - Limiters: ±5000 kW per unit
 - Outputs: UC1_P_cmd, UC2_P_cmd, UC1_Q_cmd, UC2_Q_cmd
- Section 2: ISLANDING DETECTION
 - Inputs: df/dt, V_phase_angle, Utility_BRK
 - ABS + Comparator: |df/dt| > 1.5 Hz/s (ROCOF)
 - Delay + Summer + ABS + Comparator: Δangle > 10° (vector surge)
 - NOT gate: breaker open detection
 - OR gate: combine detections
 - Integrator + Comparator: 100ms confirmation timer
 - SR Flip-Flop: island latch
 - Output: ISLAND_DETECTED
- Section 3: RESYNCHRONIZATION
 - Inputs: V_grid, V_island, f_grid, f_island, phi_grid, phi_island
 - 3 × (Summer + ABS + Comparator): ΔV<3%, Δf<0.1Hz, Δφ<10°
 - AND gate: all conditions met
 - Integrator: 1s hold timer
 - Comparator: hold time exceeded → SYNC_READY
 - Integrator + Comparator: 30s timeout → SYNC_FAIL
 - Outputs: SYNC_READY, SYNC_FAIL
- Section 4: LOAD SHEDDING
 - Inputs: f_grid, SOC_avg
 - 4 × Comparators: f < 59.5/59.3/59.0/58.5 Hz
 - 4 × Delay blocks: 0.5s shed confirmation
 - 4 × SR Flip-Flops: tier latch
 - Outputs: SHED_TIER1..4 (breaker commands)
- Section 5: BLACK-START SEQUENCER
 - Inputs: BS_TRIGGER, DC_OK, MCB_closed, V_bus_ok, GT_ready
 - 8 × SR Flip-Flops: step states (0-7)
 - AND gates: step transition conditions
 - Output: BS_STEP (current step number)
- Section 6: STATE MACHINE
 - 9 × SR Flip-Flops: INIT, STANDBY, GRID_TIED, ISLAND_DETECT, ISLANDED, RESYNC, RECONNECT, BLACK_START, EMERGENCY
 - AND/OR gates: transition logic
 - Output: MGC_State (encoded)
- Section 7: MONITORING
 - Graph channels: P_plant, MGC_State, SOC_avg, Island, LoadShed

5.2 Key Configurable Parameters

Category	Parameter	Default	Description
Islanding	ROCOF_TH	1.5 Hz/s	ROCOF trip threshold
	VS_TH	10.0°	Vector surge trip threshold
	ISLAND_CONFIRM	0.1 s	Confirmation time
Resync	SYNC_DV	0.03 pu	Voltage tolerance
	SYNC_DF	0.1 Hz	Frequency tolerance
	SYNC_DPHI	10.0°	Phase tolerance
	SYNC_HOLD	1.0 s	Hold time before close
	SYNC_TIMEOUT	30.0 s	Max resync attempt time
Load Shed	FREQ_SHED_1..4	59.5/59.3/59.0/58.5 Hz	Tier thresholds
	SOC_SHED_1	20.0%	SOC-based shedding
	SHED_CONFIRM	0.5 s	Shed confirmation delay
	RESTORE_CONFIRM	5.0 s	Restore confirmation delay

6. EMS/SCADA Implementation

6.1 PSCX Block Architecture

```

EMS_SCADA Page Module (Multi-Rate Factor = 20000)
├── Section 1: TOU SCHEDULER
│   ├── Inputs: TOU_period, SOC_avg
│   ├── Selector (3-input): Off-peak/Mid-peak/On-peak
│   ├── Comparators: SOC vs charge target (90%) and discharge min (20%)
│   ├── Multipliers: gate rate by SOC condition
│   └── Output: P_tou (-5000 / 0 / +5000 kW)
├── Section 2: PEAK-SHAVING PI CONTROLLER
│   ├── Inputs: Load_P, P_plant_meas, SOC_avg
│   ├── Summer: P_utility = Load - P_plant
│   ├── Hysteresis: activation at 4200 kW, deactivation at 3800 kW
│   ├── SR Flip-Flop: activation latch
│   ├── PI Controller: error = P_util - 4000, Kp=1.0, Ki=0.1
│   ├── SOC gate: only active if SOC > 25%
│   └── Output: P_peak_shave, PS_active
├── Section 3: DEMAND RESPONSE
│   ├── Inputs: VPP_signal, VPP_active
│   ├── Multiplier: gate signal by active flag
│   └── Output: P_dr, DR_flag
├── Section 4: PRIORITY ARBITRATION
│   ├── Inputs: MGC_Fault, all dispatch sources, plant availability
│   ├── Safety comparator: critical fault check
│   ├── 3-level selector chain: Safety > DR > Peak-shave+TOU
│   ├── Limiter: ±10000 kW plant range
│   └── Output: P_dispatch, PlantMode
└── Section 5: MONITORING
    └── Graph channels: P_dispatch, P_tou, P_ps, TOU_period, DR_flag
  
```

6.2 Dispatch Schedule File Format

The file `Resources/dispatch_schedule.txt` uses PSCAD File Read format:

! Time(s)	P_dispatch(kW)	Q_dispatch(kVAR)	TOU_period	PlantMode
0.0	-5000	0	0	1
7.0	0	0	1	1
14.0	5000	0	2	1
22.0	-5000	0	0	1

Time mapping: 1 simulation second = 1 real hour (configurable via TIME_SCALE constant)

To modify the schedule:

1. Edit `dispatch_schedule.txt` with any text editor
2. Follow the column format exactly
3. Ensure times are monotonically increasing
4. Set simulation duration $\geq 24s$ for full cycle

7. System Integration

7.1 Page Module Hierarchy

```

PSCAD Project
├── Main Page (existing – DS Station Definition)
│   ├── EMS_SCADA (new page module)
│   │   ├── Multi-Rate Factor: 20000 (1 s execution)
│   │   └── Contains: EMS_SCADA component instance
│   ├── MGC_Controller (new page module)
│   │   ├── Multi-Rate Factor: 200 (10 ms execution)
│   │   └── Contains: MGC_Controller component instance
│   ├── UC_1 (new page module)
│   │   ├── Multi-Rate Factor: 20 (1 ms execution)
│   │   └── Contains: UC_Controller component instance #1
│   ├── UC_2 (new page module)
│   │   ├── Multi-Rate Factor: 20 (1 ms execution)
│   │   └── Contains: UC_Controller component instance #2
│   ├── BESS_1_1 (existing – PCS+BESS Unit 1, UNMODIFIED)
│   ├── BESS_1_2 (duplicate of BESS_1_1 for Unit 2)
│   ├── Source_1 (existing – AC grid equivalent)
│   └── Multimeter (existing – POI measurements)

```

7.2 Connecting to the Existing PCS Black Box

The existing `BESS_1_1` subsystem has these key interfaces:

PCS Interface	Connection Method	Notes
PQref (2-dim input)	Replace existing Pset / Qset sliders with UC outputs via datamerge	The existing datamerge block at (108, 432) on Main combines P and Q
MCB_DRV1 (breaker name)	UC MCB_DRV output → breaker control signal	Already named in BESS_1_1
Slider switches (AVR, FW_EN, etc.)	UC outputs → slider control overrides	Use PSCAD “control from signal” feature
P , Q measurements	FFT_Mod_2 outputs → UC inputs via data labels	Read existing output channels
BatSOC	BESS model output → UC SOC input	Available on BESS_1_1 outputs
Batvolt	BESS model output → UC V_bat input	Available on BESS_1_1 outputs
Vdc_meas	DC link measurement → UC V_dc input	Already labeled in Main
Vlrms	FFT line voltage RMS → UC V_ac_rms input	3-dimensional signal
PCS Fault	Ctrl_1_2 fault outputs → UC PCS_Fault input	4-dimensional bitmask

Critical Wiring Changes on Main Page:

1. **Disconnect** the Pset slider (master:var, id=1905569060) from the datamerge input
2. **Disconnect** the Qset slider (master:var, id=683095525) from the datamerge input
3. **Connect** data label UC1_PCS1_P_ref to datamerge input 1
4. **Connect** data label UC1_PCS1_Q_ref to datamerge input 2
5. Keep Pset and Qset sliders for manual testing mode (use a selector switch)

7.3 Duplicate PCS for Unit 2

To create the second PCS+BESS unit:

1. **Copy** the entire BESS_1_1 definition
2. **Rename** to BESS_1_2
3. **Rename** all internal labels to avoid conflicts (suffix with _2)
4. Place BESS_1_2 on the Main page
5. Connect UC_2 outputs to BESS_1_2 inputs using data labels with UC2_PCS2_ prefix

8. Signal Routing & Data Labels

8.1 Naming Convention

All cross-page data labels follow:

```
<SOURCE>_<DESTINATION>_<SIGNAL_NAME>
```

Examples:

Label Name	Direction	Description
EMS_MGC_P_dispatch	EMS → MGC	Active power dispatch command
MGC_UC1_P_cmd	MGC → UC1	Unit 1 active power command
MGC_UC2_P_cmd	MGC → UC2	Unit 2 active power command
UC1_MGC_SOC	UC1 → MGC	Unit 1 SOC feedback
UC1_PCS1_P_ref	UC1 → PCS1	Unit 1 P reference to PCS
PCS1_UC1_P_meas	PCS1 → UC1	PCS1 measured P to UC1
MGC_EMS_P_plant	MGC → EMS	Plant aggregate P measurement
GRID_f_grid	Grid → MGC	Grid frequency measurement
GRID_df_dt	Grid → MGC	Rate of change of frequency

8.2 Complete Signal List

EMS ↔ MGC (5 down / 14 up)

#	Label	Dir	Type	Dim	Update
1	EMS_MGC_P_dispatch	EMS→MGC	Real	1	1 s
2	EMS_MGC_Q_dispatch	EMS→MGC	Real	1	1 s
3	EMS_MGC_PlanMode	EMS→MGC	Int	1	1 s
4	EMS_MGC_DR_flag	EMS→MGC	Int	1	1 s
5	EMS_MGC_EMS_heartbeat	EMS→MGC	Int	1	1 s
6	MGC_EMS_P_plant_meas	MGC→EMS	Real	1	100 ms
7	MGC_EMS_Q_plant_meas	MGC→EMS	Real	1	100 ms
8	MGC_EMS_SOC_avg	MGC→EMS	Real	1	100 ms
9	MGC_EMS_SOC_min	MGC→EMS	Real	1	100 ms
10	MGC_EMS_MGC_State	MGC→EMS	Int	1	100 ms
11	MGC_EMS_MGC_Status	MGC→EMS	Int	1	100 ms
12	MGC_EMS_MGC_Fault	MGC→EMS	Int	1	100 ms
13	MGC_EMS_P_plant_avail	MGC→EMS	Real	1	100 ms
14	MGC_EMS_P_plant_chg	MGC→EMS	Real	1	100 ms
15	MGC_EMS_N_units_online	MGC→EMS	Int	1	100 ms
16	MGC_EMS_MGC_heartbeat	MGC→EMS	Int	1	100 ms

MGC ↔ UC (7×2 down / 11×2 up)

#	Label Pattern	Dir	Type	Dim
1	MGC_UC{n}_P_cmd	MGC→UC	Real	1
2	MGC_UC{n}_Q_cmd	MGC→UC	Real	1
3	MGC_UC{n}_Mode_cmd	MGC→UC	Int	1
4	MGC_UC{n}_Enable	MGC→UC	Int	1
5	MGC_UC{n}_Ramp_limit	MGC→UC	Real	1
6	MGC_UC{n}_V_dc_ref	MGC→UC	Real	1
7	MGC_UC{n}_MGC_heartbeat	MGC→UC	Int	1
8	UC{n}_MGC_P_meas	UC→MGC	Real	1
9	UC{n}_MGC_Q_meas	UC→MGC	Real	1
10	UC{n}_MGC_SOC	UC→MGC	Real	1
11	UC{n}_MGC_V_ac_rms	UC→MGC	Real	1
12	UC{n}_MGC_V_dc	UC→MGC	Real	1
13	UC{n}_MGC_UC_Status	UC→MGC	Int	1
14	UC{n}_MGC_UC_Fault	UC→MGC	Int	1
15	UC{n}_MGC_UC_Mode	UC→MGC	Int	1
16	UC{n}_MGC_P_availability	UC→MGC	Real	1
17		UC→MGC	Real	1

#	Label Pattern	Dir	Type	Dim
	UC{n} _MGC_P_charge_a vail			
18	UC{n} _MGC_UC_heartbe at	UC→MGC	Int	1

Where {n} = 1 or 2 for each unit.

9. Multi-Rate Configuration

9.1 Setting Multi-Rate Factor in PSCAD

1. Double-click the **Page Module** header in the project tree
2. In the Properties dialog, find **Multi-Rate Factor**
3. Enter the value per the table below
4. Click OK

Page Module	Multi-Rate Factor	Effective Rate	Base Δt
Main / BESS_1_1	1 (default)	50 μs	50 μs
UC_1	20	1.0 ms	50 μs × 20
UC_2	20	1.0 ms	50 μs × 20
MGC	200	10 ms	50 μs × 200
EMS	20000	1.0 s	50 μs × 20000

9.2 Inter-Rate Signal Handling

When signals cross multi-rate boundaries, PSCAD automatically samples them. However:

- **Fast→Slow** (e.g., PCS at 50μs → UC at 1ms): PSCAD uses the most recent value. No special handling needed.
- **Slow→Fast** (e.g., EMS at 1s → MGC at 10ms): The signal holds its last value between updates. This is correct — dispatch commands are piecewise-constant.
- **Sampler blocks** are included in the graphical implementation at rate boundaries for explicit control.

9.3 Timing Verification

After building, verify timing in the simulation output:

1. Monitor UC_heartbeat — should increment every 1 ms (1000/s)
2. Monitor MGC_heartbeat — should increment every 10 ms (100/s)

3. Monitor `EMS_heartbeat` — should increment every 1 s (1/s)

10. Test Procedures

10.1 Unit Controller Tests

UC-T01: Ramp Limiter Verification

Step	Action	Expected Result
1	Set Enable=1, Mode_cmd=1	UC enters PRECHARGE → CONNECTING → RUNNING
2	Step P_cmd from 0 to 5000 kW	P_ref ramps at 833.33 kW/s (reaches 5000 kW in ~6s)
3	Verify ramp rate on graph	$dP/dt \leq 833.33 \text{ kW/s}$ at all times
4	Step P_cmd from 5000 to -5000 kW	P_ref ramps down then up at same rate

UC-T02: SOC Floor Protection

Step	Action	Expected Result
1	Set SOC=50%, P_cmd=5000 kW	P_ref = 5000 kW (no clamping)
2	Reduce SOC to 12%	P_ref derated to 40% = 2000 kW
3	Reduce SOC to 10%	P_ref = 0 kW (hard stop)
4	Try charging at SOC=10%	P_ref follows Q_cmd (charge OK)

UC-T03: SOC Ceiling Protection

Step	Action	Expected Result
1	Set SOC=50%, P_cmd=-5000 kW	P_ref = -5000 kW (charging)
2	Increase SOC to 92%	P_ref derated to 60% = -3000 kW
3	Increase SOC to 95%	P_ref = 0 kW (charge blocked)
4	Try discharging at SOC=95%	P_ref follows P_cmd (discharge OK)

UC-T04: Fault Response

Step	Action	Expected Result
1	Set PCS_Fault[0]=1	P_ref=0, Q_ref=0, MCB opens within 2ms
2	Clear PCS_Fault[0]=0	UC stays in FAULTED state
3	Wait 5 seconds	UC transitions to FAULT_CLEAR → CONNECTING
4	Verify MCB closes	Returns to RUNNING state

UC-T05: Heartbeat Timeout

Step	Action	Expected Result
1	Normal operation with MGC_heartbeat incrementing	UC RUNNING normally
2	Freeze MGC_heartbeat (stop incrementing)	After 500ms: COMM_LOSS fault
3	Verify P_ref ramps to zero	Ramp-down, not instant cutoff
4	Resume MGC_heartbeat	COMM_LOSS clears automatically

10.2 Microgrid Controller Tests

MGC-T01: SOC-Weighted Sharing (Equal)

Input	Value
SOC_1	50%
SOC_2	50%
P_dispatch	6000 kW

Expected Output	Value
UC1_P_cmd	3000 kW
UC2_P_cmd	3000 kW

MGC-T02: SOC-Weighted Sharing (Unequal Discharge)

Input	Value
SOC_1	80%
SOC_2	40%
P_dispatch	6000 kW

Expected Output	Value
UC1_P_cmd	4000 kW $(80/(80+40) \times 6000)$
UC2_P_cmd	2000 kW $(40/(80+40) \times 6000)$

MGC-T03: Islanding Detection

Step	Action	Expected Result
1	Normal grid-tied operation	MGC_State = GRID_TIED
2	Inject $df/dt = 2.0$ Hz/s	ROCOF trip detected
3	Wait 100ms	ISLAND_DETECT → ISLANDED
4	Verify lead UC → GFM	Higher-SOC unit gets Mode_cmd=2
5	Verify utility breaker command	Utility_BRK_cmd = 0 (open)

MGC-T04: Load Shedding

Step	Action	Expected Result
1	Set f_grid = 59.4 Hz	Below FREQ_SHED_1 (59.5 Hz)
2	Wait 0.5s (confirmation)	Tier 4 load shed (Load-Shed_cmd_4 = 0)
3	Set f_grid = 59.2 Hz	Below FREQ_SHED_2 (59.3 Hz)
4	Wait 0.5s	Tier 3 shed added
5	Restore f_grid = 60.0 Hz	Tiers restore after 5s each

10.3 EMS/SCADA Tests**EMS-T01: TOU Schedule**

Step	Action	Expected Result
1	Set TOU_period=0 (off-peak), SOC=50%	P_dispatch = -5000 kW (charging)
2	Set TOU_period=1 (mid-peak)	P_dispatch = 0 kW (standby)
3	Set TOU_period=2 (on-peak), SOC=50%	P_dispatch = +5000 kW (discharge)
4	Set TOU_period=2, SOC=18%	P_dispatch = 0 kW (below minimum)

EMS-T02: Peak-Shaving

Step	Action	Expected Result
1	Set Load_P=3000 kW	No peak-shaving (below 4000 kW)
2	Set Load_P=5000 kW	Peak-shave activates, BESS discharges to cap import at 4000 kW
3	Verify: P_dispatch $\approx$ 1000 kW	Import capped: 5000 - 1000 = 4000 kW
4	Set Load_P=3500 kW	Peak-shave deactivates (below 3800 kW threshold)

EMS-T03: DR Override

Step	Action	Expected Result
1	Set VPP_active=0, TOU_period=0	Normal TOU charging
2	Set VPP_active=1, VPP_signal=8000	P_dispatch = 8000 kW (DR overrides TOU)
3	Set VPP_active=0	Returns to TOU schedule

10.4 Integration Test: Full Dispatch Chain

Step	Time	Action	Verify
1	t=0s	Start sim, TOU=off-peak	EMS: -5000kW → MGC splits → UC1/UC2 charge
2	t=7s	TOU→mid-peak	EMS: 0kW → system holds
3	t=14s	TOU→on-peak	EMS: +5000kW → system discharges
4	t=15s	Inject island event	MGC: GRID_TIED → ISLAND_DETECT → IS-LANDED
5	t=16s	Verify GFM operation	Lead UC in GFM, load shedding if needed
6	t=20s	Grid restores	MGC: RESYNC → RE-CONNECT → GRID_TIED
7	t=22s	TOU→off-peak	EMS: charge cycle begins

11. Troubleshooting

11.1 Common Issues

Issue	Cause	Solution
“Undefined data label” error	Data label name mismatch between pages	Check spelling exactly matches §8 convention
Multi-rate factor not applying	Page module not configured	Double-click page module properties, set Multi-Rate Factor
P_ref stays at zero	UC state machine stuck in IDLE	Verify Enable=1 and Mode_cmd>0 are reaching UC
SOC derating not working	Wrong derating gain value	Verify gain = $1 / (\text{SOC_LOW_WARN} - \text{SOC_FLOOR}) = 0.2$
Islanding not detected	ROCOF signal not computed	Add derivative block (df/dt) on frequency measurement
Peak-shaving oscillation	PI gains too high	Reduce PS_KI to 0.05 or PS_KP to 0.5
Fortran compilation error	Array dimension mismatch	Ensure V_AC_RMS declared as REAL(3) and PCS_FAULT as INTEGER(4)
Simulation very slow	EMS Multi-Rate too small	Ensure EMS uses 20000, not 200
BESS_1_1 behavior changes	Modified internal components	NEVER modify BESS_1_1 internals — it is a validated black box

11.2 Signal Debugging

To trace signal flow issues:

1. **Add output channels** (master:pgb) to any signal wire
2. **Use Digital Meters** (master:meter) for real-time value display
3. **Add data labels** as temporary probes — set Conv to “input” for read-only monitoring
4. **Check heartbeat counters** to verify controller execution rates

11.3 Performance Optimization

If simulation runs too slowly:

Setting	Current	Optimization
Time duration	25s	Reduce to 1s for quick tests
Time step	50 μ s	Cannot increase (PCS requires it)
UC multi-rate	20	Can increase to 40 (2ms) for initial testing
MGC multi-rate	200	Can increase to 400 (20ms) for initial testing
EMS multi-rate	20000	Already optimal

11.4 Version Compatibility

- These files target **PSCAD V5.0.2** specifically
 - The XML schema uses V5.0.2 classid and attribute conventions
 - If using a different PSCAD version, the `generate_pscx.py` script can be modified to target different schemas
 - The Fortran files (`.f`) are standard Fortran 77 and should compile with any PSCAD-supported compiler
-

Appendix A: Algorithm Cross-Reference

Algorithm ID	Design Doc §	PSCX Section	Fortran Function	Test ID
ALG-01	§2.5.1	UC Section 1	UC_CONTROLLER lines 120-145	UC-T01
ALG-02	§2.5.2	UC Section 2	UC_CONTROLLER lines 150-185	UC-T02, UC-T03
ALG-03	§2.5.3	UC Section 3	UC_CONTROLLER lines 50-100	UC-T04, UC-T05
ALG-04	§2.5.4	UC Section 5	UC_CONTROLLER lines 105-130	—
ALG-05	§2.5.5	UC Section 6	UC_CONTROLLER lines 190-215	—
ALG-06	§3.5.1	MGC Section 1	MGC_CONTROLLER lines 80-130	MGC-T01, T02
ALG-07	§3.5.2	MGC Section 2	MGC_CONTROLLER lines 40-70	MGC-T03
ALG-08	§3.5.3	MGC Section 3	MGC_CONTROLLER lines 135-155	—
ALG-09	§3.5.4	MGC Section 4	MGC_CONTROLLER lines 165-195	MGC-T04
ALG-10	§3.5.5	MGC Section 5	MGC_CONTROLLER lines 145-170	—
ALG-11	§4.4.1	EMS Section 2	EMS_SCADA lines 60-90	EMS-T02
ALG-12	§4.4.2	EMS Section 1	EMS_SCADA lines 35-55	EMS-T01
ALG-13	§4.4.3	EMS Section 3	EMS_SCADA lines 95-105	EMS-T03
ALG-14	§4.4.4	EMS Section 4	EMS_SCADA lines 110-140	—

Appendix B: File Checksums

Generated files can be verified with:

```
md5sum BESS_3Tier_Controller.pscx  
md5sum Subsystems/*.pscx  
md5sum Resources/*.f
```

End of PSCAD Implementation Guide